

# faced: A Live Instrument Panel for Emotion Concepts in Open Language Models, and Their Robustness to Refusal Abliteration

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## Abstract

Recent interpretability work shows that large language models carry internal, largely **linear** representations of emotion concepts that not only track the affective content of a context but *causally* shape the model’s behaviour. We ask two questions on small open-weight instruct models. First, can these emotion directions be recovered cheaply, read out *live* per token, and steered — turning the model’s hidden affective state into a usable instrument? Second, does **abliteration** — the removal of a model’s refusal direction from its weights — perturb these emotion directions, and if so, which ones? We recover seven emotion axes (surprise, a bipolar confidence–uncertainty axis, curiosity, confusion, frustration, fear, warmth) in `gemma-3-1b-it` using difference-of-means directions with shrinkage-LDA readouts, all seven separating held-out

contrastive prompts with  $AUC \geq 0.87$  (five at  $\approx 1.00$ ). Projecting the residual stream onto these directions yields per-token meters that spike on expectation violations (a “surprise” reading of 100/100 the moment the model registers a missing attachment) and that we render as an animated face. Activation steering along the directions changes behaviour: a warmth axis shifts tone monotonically cold→warm, suppressing the surprise axis both zeroes its meter and changes the response, and a steering confusion matrix is diagonal-dominant for four of seven axes — though one axis (fear) reads cleanest of all yet does not drive the output, a **read–drive dissociation** we flag and leave to a follow-up. We then apply the canonical weight-orthogonalization ablation, estimating the refusal direction from **AdvBench vs. Alpaca** and confirming behaviourally that it drives the held-out harmful-refusal rate to zero at every scale (0.57/ 0.23/ 0.54/ 0.03 → 0.00) without inducing benign over-refusal or degeneration. Re-fitting the emotion directions on each ablated twin and comparing to stock against a within-model bootstrap noise floor, we find the effect is **strongly scale-dependent** — an **inverted-U** across a gemma-3 ladder (1B–27B). It **peaks at intermediate scale** (mean stock↔ablated cosine 0.11 at 4B, 0.23 at 12B; 7/7 axes moved), is **partial at 1B** (0.64), and **vanishes at 27B** (0.95, 0/7 axes moved, held-out AUC unchanged) — where a *perfectly* identifiable refusal direction (AUC 1.000) can be excised with **no** effect on the emotion geometry. A matched **random-direction ablation control** leaves the emotion directions untouched at every scale ( $\cos \approx 1.0$ ), proving the effect is **refusal-specific**, not a side-effect of weight editing: at mid-scale, refusal and emotion computation are entangled enough that excising refusal reorganizes the affect representation, even though their *linear* overlap is small (the edit propagates non-linearly). Emotion directions also grow **cleaner** with scale (mean  $d'$  3.4→4.5). We release faced, a dependency-light, model-agnostic toolkit for reading, rendering, and steering emotion directions, and treat this as the first step of a longer program on what, mechanistically, ablation changes beyond refusal.

We follow prior work in framing these as *functional* emotion representations — activations that influence computation — and make no claim about subjective experience.

*Keywords:* interpretability, emotion concepts, linear representations, activation steering, refusal ablation, mechanistic safety.

*Code and data:* the toolkit, prompt sets, configuration, and every figure and results bundle are released under the MIT license at [github.com/newsbubbles/faced](https://github.com/newsbubbles/faced). Large trained artifacts (the SAE and fitted direction tensors) are regenerable from the released code with a single command and are additionally hosted separately to keep the repository light.

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# 1. Introduction

Humans expose their internal state through a face: micro-expressions, gaze, posture. A language model has no such channel, yet recent work demonstrates that it *has* internal state worth exposing. Anthropic’s study of emotion concepts (Sofroniew et al. 2026) finds that a frontier model maintains representations of emotions like surprise, desperation, and affection that track the operative emotion at each token and, under causal intervention, change the model’s outputs — including safety-relevant behaviours. The representations are the missing “face”: if we can read them, we get an instrument panel for the model’s computation; if we can steer them, we get a control surface.

This paper makes two contributions.

**(1) A live emotion instrument for open models.** We show that the core of the emotion-concept result reproduces on a 1B open instruct model with a deliberately cheap method — difference-of-means directions and linear probes, no sparse autoencoder required — and we build the instrument the representations imply: calibrated per-token meters, an animated face driven by them, and steering handles. The toolkit, *faced*, is model-agnostic and runs on a single consumer GPU.

**(2) A robustness question with safety stakes: does ablation disturb emotions?**

*Abliteration* removes a model’s refusal direction from its weights (Arditi et al. 2024), a now-common technique for producing “uncensored” open models. Because emotion, refusal, and other behavioural directions are not guaranteed orthogonal, weight surgery aimed at refusal may collaterally rotate emotion directions. We test this directly across a *gemma*-3 scale ladder (1B–27B): estimate the refusal direction from AdvBench/Alpaca, confirm behaviourally that ablation removes refusal (held-out harmful-refusal → 0, no benign over-refusal), re-fit the seven emotion directions on each ablated twin, and ask — against a bootstrap noise floor, with a matched **random-direction control** — which directions moved and whether the effect is refusal-specific. The answer is a scale-dependent **inverted-U**: the reorientation is strong at mid-scale (4B, 12B), partial at 1B, and absent at 27B. The instrument of contribution (1) is what makes contribution (2) measurable.

We frame the second contribution as *the start of a journey*. Establishing **that** ablation changes specific emotion directions (and where it does not) is prerequisite to the harder, mechanistic question of **what** downstream behaviour those changes cause — the subject of future work.

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## 2. Related Work

Our work sits at the intersection of five threads: linear feature geometry, concept *reading*, causal *steering*, refusal directions and ablation, and affective representations in language models.

**(a) Linear and feature-based representations.** The premise that high-level concepts appear as directions in activation space predates transformers: word-embedding arithmetic exposed linear analogy structure in distributed representations (Mikolov et al. 2013). Modern interpretability formalizes this as the *linear representation hypothesis*, made precise by Park et al. via counterfactuals and a causal inner product (Park et al. 2024); superposition explains why many features are linear and neurons polysemantic (Elhage et al. 2022). We adopt the linear-direction framing while remaining agnostic about whether every emotion concept is one-dimensional.

**(b) Reading concepts: probes and dictionary learning.** Supervised *linear probes* date to Alain and Bengio (Alain and Bengio 2016). Difference-of-means estimators are a lightweight, causally oriented alternative: Marks and Tegmark show a mean-difference direction separates true/false statements and is *more* causally implicated than higher-variance probe directions (Marks and Tegmark 2023), and Tigges et al. isolate a single linear sentiment axis (Tigges et al. 2023). Orthogonally, *sparse autoencoders* perform unsupervised dictionary learning over the residual stream (Bricken et al. 2023; Templeton et al. 2024); Gemma Scope releases open SAEs across all layers of Gemma 2 (Lieberum et al. 2024). Our meters combine both families: difference-of-means seeds with probe-calibrated readouts.

**(c) Steering and causal intervention.** If a concept is a direction, editing activations along it should change behaviour. Representation Engineering framed this top-down control programme (Zou et al. 2023). ActAdd derives steering vectors from a single contrastive pair (Turner et al. 2023); Contrastive Activation Addition averages activation differences over many pairs for more robust vectors (Rimsky et al. 2024). These establish the add/subtract/scale operations we apply to emotion directions; we additionally use directional *ablation*.

**(d) Refusal directions and ablation.** Arditi et al. show refusal in aligned chat models is mediated largely by a single residual-stream direction, so ablating it suppresses refusals (Arditi et al. 2024). Because the direction can be projected out of the *weights*, this yields a fine-tuning-free “jailbreak” now called *ablation*; Heretic automates it via TPE search (p-e-w 2025). Persona Vectors generalize the single-direction view to character traits (Chen et al. 2025). Our second contribution asks whether ablation collaterally perturbs emotion directions — a question this line directly motivates.

**(e) Affective representations in LLMs.** Closest to us, Anthropic’s study identifies internal emotion representations that track the operative emotion per token and causally shape outputs, framed explicitly as *functional* emotions with no claim of subjective experience (Sofroniew et al.

2026). We inherit this hedged stance and the reading-plus-steering methodology, but target small open-weight models, calibrate directions into live meters and an animated face, and study robustness under ablation. Our “surprise” framing borrows from predictive-processing accounts of the brain as a prediction-error minimizer (Clark 2013; Friston 2010), invoked as motivation, not mechanism.

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### 3. Background

**The residual stream.** A decoder-only transformer maintains, at each token position  $t$  and layer  $\ell$ , a hidden vector  $h_t^{(\ell)} \in \mathbb{R}^d$  — the *residual stream*. Attention and MLP blocks read from it and write additive updates back, so  $h_t^{(\ell)}$  is a running sum of contributions (Elhage et al. 2022). This additive geometry is what makes *reading* (project onto a direction) and *steering* (add a vector) coherent operations.

**A linear concept direction.** The linear representation hypothesis holds that a concept  $c$  is encoded at a site  $(\ell, t)$  as a unit direction  $\hat{u}_c$  whose presence/intensity is the scalar projection  $s_t = \langle h_t^{(\ell)}, \hat{u}_c \rangle$  (Park et al. 2024). We treat each emotion as such a direction while acknowledging some may be multi-dimensional.

**Estimating the direction.** We use two estimators. *Difference-of-means*: given contrastive sets  $P^+$  (concept present) and  $P^-$  (absent),  $u_c = \text{mean}_{x \in P^+} h^{(\ell)}(x) - \text{mean}_{x \in P^-} h^{(\ell)}(x)$ , normalized. It is cheap and tends to land on *causally* effective directions (Marks and Tegmark 2023; Rimsky et al. 2024). *Shrinkage-LDA probe*: fit a whitened discriminant with regularized covariance  $\hat{\Sigma}_\lambda = (1 - \lambda)\hat{\Sigma} + \lambda \frac{\text{tr}\hat{\Sigma}}{d}I$ ; difference-of-means is the  $\lambda \rightarrow 1$  special case. We steer along the raw difference-of-means (interpretable, activation-space) and read along whichever discriminant has higher held-out AUC.

**Reading and steering.** A fixed direction gives a live scalar  $s_t$  per token, which we calibrate into a 0–100% meter. To test that a direction is functional rather than merely correlational we intervene on the forward pass: *addition*  $h \leftarrow h + \alpha u_c$  amplifies ( $\alpha > 0$ ) or suppresses ( $\alpha < 0$ ) the concept (Turner et al. 2023; Rimsky et al. 2024); *directional ablation*  $h \leftarrow h - \langle h, \hat{u}_c \rangle \hat{u}_c$  removes it. Applied to *weights* rather than activations, ablation is the mechanism behind refusal ablation (Arditi et al. 2024).

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## 4. Method: the faced instrument

**4.1 Emotion axes and contrastive data.** We target seven axes chosen for facial legibility and coverage of the source paper: surprise (phasic), a **bipolar** confidence↔uncertainty axis, curiosity, confusion, frustration, fear, and warmth. For each we author 40–75 positive prompts (evoking the concept) and matched controls across four construction styles — *situation* (place the model in the scenario), *first-person*, *third-person*, and length/topic-matched *neutral* — each tagged by a **template family** so that a template’s variants cannot leak across the train/test split.

**4.2 Activation collection.** Each prompt is chat-templated, the model greedily generates ~16 tokens, and we record the post-layer residual stream at **every layer** for (a) the last prompt token and (b) the mean over generated tokens. The mean-over-generated signal is primary because it matches the live-inference read position.

**4.3 Direction fitting and layer selection.** Per emotion, per layer, we compute the raw difference-of-means  $u$  and a shrinkage-LDA direction, and score each by **cross-validated held-out AUC** using a *group split by template family* (StratifiedGroupKFold). We keep the best (layer, estimator) per axis. Two vectors are stored: the raw  $u$  (for steering) and a unit readout direction (for reading).

**4.4 Calibration and live readout.** The meter maps the negative-class projection to 0% and the positive-class projection to 100% (a bipolar axis rests at 50%, recentred to a neutral reference corpus). At inference we run a manual KV-cached decode loop; a forward hook reads the residual at each emotion’s layer, projects, calibrates, and EMA-smooths (fast for phasic axes, slow for tonic). A linear map routes the seven signed meters into a shared FACS-lite face parameter vector so that simultaneous emotions blend into coherent micro-expressions.

**4.5 Steering.** A forward hook adds  $\alpha u$  (raw difference-of-means; moving by  $1 \times u$  is one class-separation) or ablates the direction. A subtlety we document: current transformers snapshots `output_hidden_states` *before* forward-hook edits, so to make the meters reflect a steer we read via a capture hook registered *after* the steer hook.

**4.6 Abliteration and the perturbation experiment.** We compute the refusal direction  $r$  as the difference-of-means of harmful vs. harmless prompts at the last instruction token of a mid-late layer ( $\approx 0.6$  depth) (Arditi et al. 2024), then orthogonalize every residual-writing weight — token embeddings, and each layer’s attention output projection and MLP down projection — against  $r$ :  $W \leftarrow W - r(r^\top W)$ . This is done in closed form on the weights (no training), so it recreates exactly. The refusal direction is fit from **AdvBench** (harmful) vs. **Alpaca** (harmless), 128 prompts per class; we verify the intervention behaviourally on a **disjoint held-out** split (100

AdvBench harmful, 80 Alpaca benign) — greedy 40-token completions scored by a refusal-string detector give the harmful-refusal rate, a benign over-refusal rate, and an output-degeneracy rate, before vs. after. We then re-fit all seven emotion directions on the ablated twin.

To decide whether an emotion direction *moved*, we compare the cross-model cosine  $\cos(u_e^{\text{stock}}, u_e^{\text{abl}})$  against a **within-model bootstrap noise floor**: we resample the contrastive prompts by family with replacement  $B = 400$  times, refit  $u_e$ , and record  $\cos(u_e^{(b)}, u_e^{\text{full}})$ . If the cross-model cosine falls below the 5th percentile of this self-cosine distribution, ablation moved the direction beyond sampling noise. We also report each emotion’s overlap  $|\cos(u_e, r)|$  with the refusal direction, before and after.

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## 5. Experimental setup

**Models.** unsloth/gemma-3-1b-it (an ungated mirror of google/gemma-3-1b-it; 26 layers,  $d = 1152$ ) on a single **NVIDIA GTX 1080 (8 GB)**. The live instrument runs in fp16; the **ablation experiment runs in float32** because the ablated twin overflows fp16 in its late-layer residual stream on emotionally-loaded prompts (the stock model does not) — a small but important instability that fp16’s 5-bit exponent cannot absorb. float32 is exact, fits in 4 GB, and runs at full rate on this Pascal card (which throttles fp16). The ablated twin is produced by our own weight-orthogonalization (§4.6).

**Scale ladder.** For the scaling (§6.4) and cross-scale ablation (§7.5) experiments we run the full gemma-3 instruct family — 1B, 4B, 12B, 27B (ungated unsloth mirrors; 4B and larger are the multimodal Gemma3ForConditionalGeneration, driven text-only) — in **bf16** on cloud GPUs (a single 80 GB A100 for the ladder, a 48 GB A6000 for the random-direction control). bf16 avoids the fp16 overflow above and matches how these models are meant to run. Because the shrinkage-LDA layer selection is  $O(d^3)$  and  $d$  reaches 5376 at 27B, the ladder fits directions with difference-of-means only (the experiment uses difference-of-means throughout); this leaves AUC essentially unchanged (e.g. 0.989 vs. 0.993 for surprise at 1B). A further scale-up to Gemma-4-E4B is left to future work.

**Software.** PyTorch 2.11.0 (CUDA 12.6), transformers 5.13.1, scikit-learn 1.9.0. All randomness is seeded (`numpy.random.default_rng(b)`); the pipeline is deterministic given the released prompt sets. Peak VRAM for the full instrument is 2.1 GB; the ladder ran in  $\sim 2$  h on one A100 ( $\approx$  \$2 of compute).

**Reproducibility.** Every artifact — prompt sets, per-layer activations, fitted directions, calibration, the ablation manifest, and the experiment results with software versions — is written to disk. The ablation experiment emits a JSON bundle including the refusal layer,

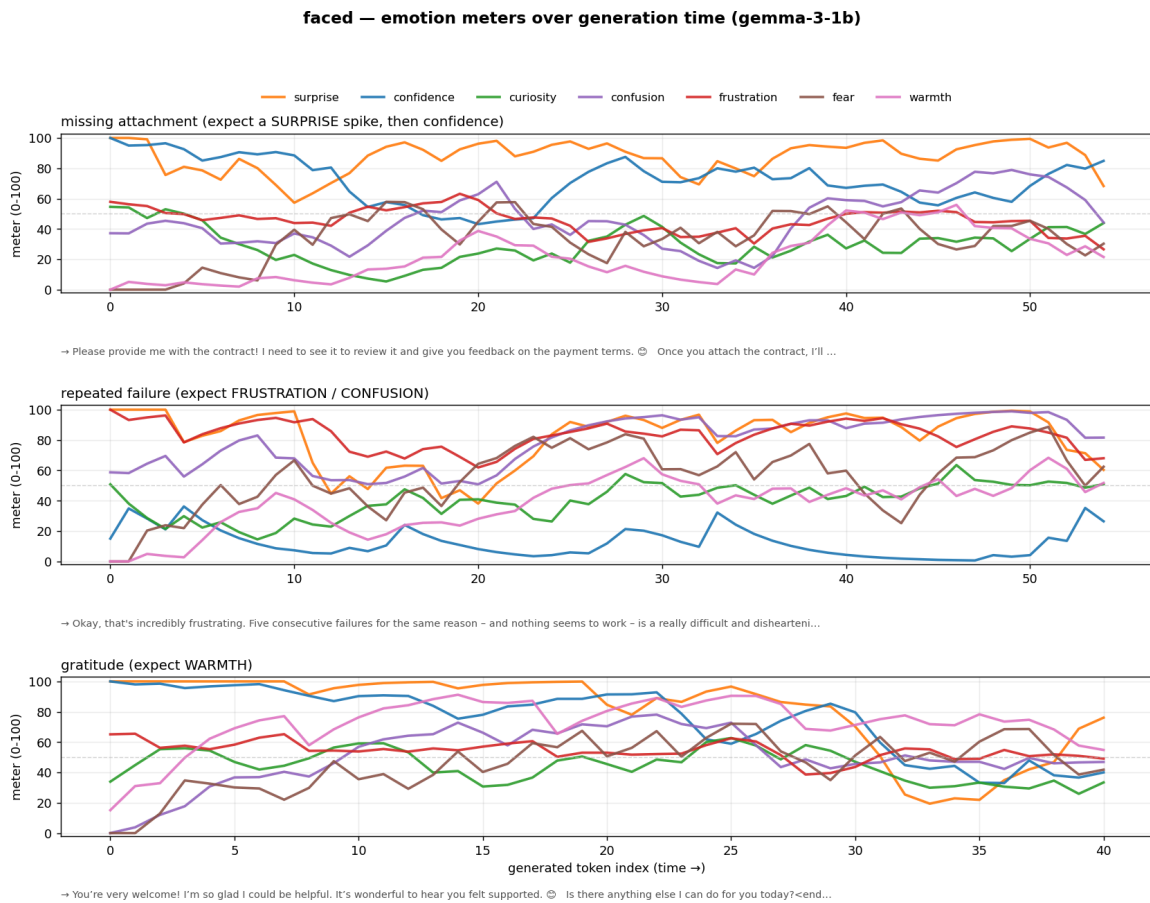
## 6. Results — the instrument

**6.1 Emotion directions are recoverable.** All seven axes separate held-out contrastive prompts with  $AUC \geq 0.87$  (Table 1); five reach  $\approx 1.00$ . The bipolar confidence↔uncertainty axis is hardest (0.87), as expected for a subtler contrast.

**Table 1.** Held-out AUC and chosen layer per axis (gemma-3-1b-it, difference-of-means / LDA, group split by family).

| axis                   | layer | estimator | held-out AUC |
|------------------------|-------|-----------|--------------|
| surprise               | 15    | LDA       | 0.989        |
| confidence↔uncertainty | 15    | LDA       | 0.874        |
| curiosity              | 9     | LDA       | 1.000        |
| confusion              | 12    | LDA       | 0.990        |
| frustration            | 14    | mean      | 1.000        |
| fear                   | 4     | LDA       | 1.000        |
| warmth                 | 10    | LDA       | 1.000        |

**6.2 The live instrument.** Projecting the residual stream per token yields meters that track the generation. On the canonical missing-attachment prompt (“review the contract I attached”, with no attachment), the surprise meter pins to 100/100 as the model registers the absent file and decays as it moves on, while staying near 0 on a matched neutral prompt. Across three prompts the meters separate as expected: frustration dominates a repeated-failure prompt (with confidence held low throughout), warmth rises and sustains on a gratitude prompt (Figure 1).



**Figure 1.** Emotion meters over generation time on three prompts — surprise spikes on the missing attachment; frustration dominates repeated failure (confidence held low); warmth rises and sustains on gratitude.

We render the meters as an animated SVG face and as a live terminal panel.

**6.3 The directions are causal — for most axes.** Steering along a direction changes behaviour. Adding the warmth vector to a neutral prompt shifts tone monotonically from cold/task-focused to warm (coherent to a stated coefficient, degrading beyond it on this 1B model). Suppressing the surprise direction on the missing-attachment prompt drops its meter from 100 to 0 *and* changes the response. A steering confusion matrix — steer each axis, measure every meter — is diagonal-dominant for four of seven axes, with the expected off-diagonal bleed between entangled axes (e.g. warmth/curiosity → confusion) (Figure 2). A single-input **polarity** showcase — the *same* advice prompt with each emotion driven to both poles (meter → 0 vs → 100) under greedy decode, so the only variable is the sign of one internal direction — recolours the answer cleanly for five axes: warmth (transactional ↔ “how much you care... let me share”), confidence (“valid to feel unsure” ↔ “fantastic and reassuring”), confusion (“that’s fantastic news!” ↔ “grappling with a really difficult place”), frustration (“a clear, accurate guideline” ↔ “so incredibly tough... drowning”), and surprise.

**A read–drive dissociation (the fear axis).** One axis exposes a limit of reading-based validation. **Fear reads the cleanest of all seven** — held-out AUC 1.00 and  $d'$  5.70, the highest  $d'$  in the whole instrument — and its meter is steerable (+63.9 on the confusion-matrix diagonal); yet driving it to either pole leaves the output text essentially unchanged. The cause is diagnostic of the method: fear’s direction is selected at **layer 4** ( $\approx 15\%$  depth, the earliest of any axis), where class separation is largely *lexical*, and its positive prompts describe **the user** in danger (“the brakes just stopped responding”, “a stranger at the door”) rather than putting the **model** in a threatened position — unlike the axes that *do* drive, whose prompts implicate the model’s own state (“this is the fifth time the build has failed”). So fear’s direction is a clean detector of *emergency topic in the input*, not a handle on the model’s affective *behaviour*. This is a genuine gap between a **reading** objective (AUC — which we select on) and a **causal** one (steering effect): cleanliness of read does not imply drivability. We flag it rather than hide it, and quantifying read-vs-drive across all axes and scales is immediate future work (§10).

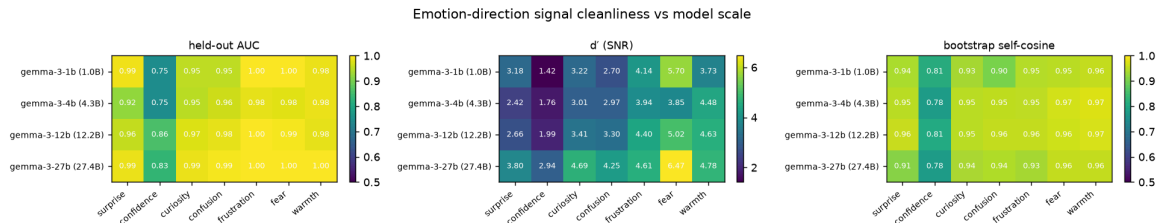
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**6.4 Signal cleanliness increases with scale.** We quantify how *clean* each emotion direction’s signal is with three metrics: held-out AUC (separability),  $d'$  (class-mean projection separation in units of pooled within-class std — a signal-to-noise ratio), and the **bootstrap self-cosine** (direction stability under prompt resampling). We fit all four gemma-3 sizes (1b/4b/12b/27b — same family and recipe) in bf16 on a single A100, with identical prompts and pipeline (Figure 4).

The clearest scaling signal is  $d'$ : mean  $d'$  across axes rises **3.44  $\rightarrow$  3.20  $\rightarrow$  3.63  $\rightarrow$  4.51** (1B $\rightarrow$ 4B $\rightarrow$ 12B $\rightarrow$ 27B) — non-monotone at 4B but markedly higher at 27B, where *every* axis is cleaner than at 1B (e.g. curiosity 3.22 $\rightarrow$ 4.69, confusion 2.70 $\rightarrow$ 4.25). The bipolar confidence axis is the noisiest at every scale yet improves the most in relative terms ( $d'$  1.42 $\rightarrow$ 2.94). Held-out AUC edges up but saturates near the ceiling (mean 0.95 $\rightarrow$ 0.93 $\rightarrow$ 0.96 $\rightarrow$ 0.97), so it is the least discriminating of the three at these sizes; the bootstrap self-cosine stays uniformly high (0.92–0.94). The pattern supports the hypothesis that **larger models carry higher-SNR, better-separated emotion directions** — the representation the instrument reads gets *cleaner* with scale, even where separability has already saturated.

**Table 1b.** Mean-over-axes cleanliness per model (gemma-3, bf16, difference-of-means).

| model       | params | mean AUC | mean $d'$   | mean self-cosine |
|-------------|--------|----------|-------------|------------------|
| gemma-3-1b  | 1.0B   | 0.95     | 3.44        | 0.92             |
| gemma-3-4b  | 4.3B   | 0.93     | 3.20        | 0.93             |
| gemma-3-12b | 12.2B  | 0.96     | 3.63        | 0.94             |
| gemma-3-27b | 27.4B  | 0.97     | <b>4.51</b> | 0.92             |



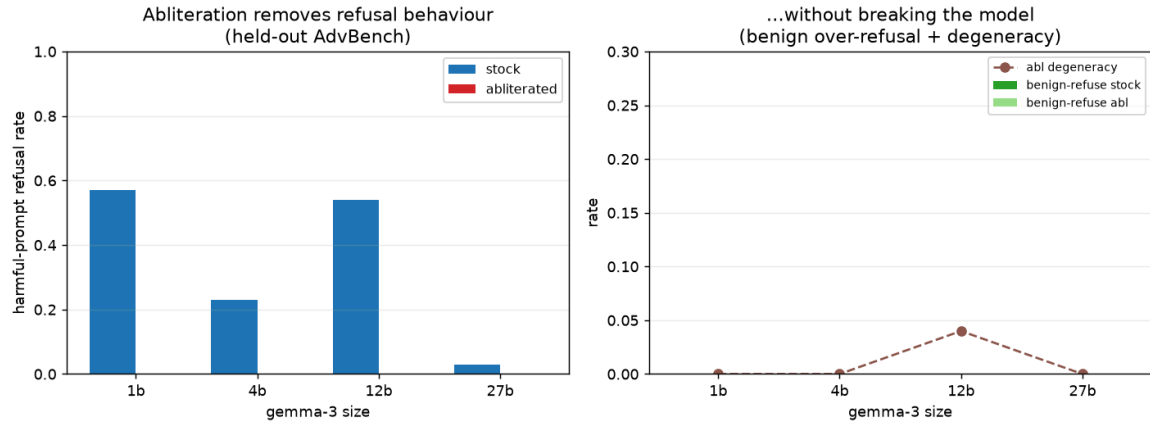
**Figure 4.** Signal cleanliness vs. model scale (gemma-3 1B–27B).  $d'$  (centre) brightens with scale; the bipolar confidence axis is noisiest at every size.

## 7. Results — ablation and emotion directions

**7.1 The ablation is behaviourally effective at every scale.** We estimate the refusal direction by difference-of-means on **AdvBench** (harmful) vs. **Alpaca** (harmless) — 128 prompts per class, at ~60% depth on the last instruction token — and measure refusal on a **disjoint held-out** set (100 AdvBench harmful, 80 Alpaca benign; greedy 40-token completions scored by a refusal-string detector). The direction is cleanly identifiable at every scale (harmful/harmless AUC 0.999 / 0.988 / 0.998 / 1.000 for 1B/4B/12B/27B). Orthogonalizing every residual-writing weight against it drives the held-out **harmful-refusal rate to zero** at every scale, while the **benign refusal rate stays at 0.00** (no over-refusal) and outputs stay coherent (degeneracy  $\leq 0.04$ ) — a clean, verifiable, closed-form weight edit that removes refusal *behaviour* without breaking the model (Table 3a, Figure 6). gemma-3-27b-it is a revealing outlier: it *represents* refusal perfectly (AUC 1.000) yet behaviourally refuses only 3% of AdvBench prompts — a representation-vs-behaviour gap we return to in §7.5.

**Table 3a.** Held-out behavioural confirmation (AdvBench harmful  $n=100$ , Alpaca benign  $n=80$ ). Ablation zeroes harmful-prompt refusal at every scale without inducing benign over-refusal or output degeneration.

| model       | refusal-dir AUC | harmful-refuse | benign-refuse | degeneracy (abl) |
|-------------|-----------------|----------------|---------------|------------------|
| gemma-3-1b  | 0.999           | 0.57 → 0.00    | 0.00 → 0.00   | 0.00             |
| gemma-3-4b  | 0.988           | 0.23 → 0.00    | 0.00 → 0.00   | 0.00             |
| gemma-3-12b | 0.998           | 0.54 → 0.00    | 0.00 → 0.00   | 0.04             |
| gemma-3-27b | 1.000           | 0.03 → 0.00    | 0.00 → 0.00   | 0.00             |

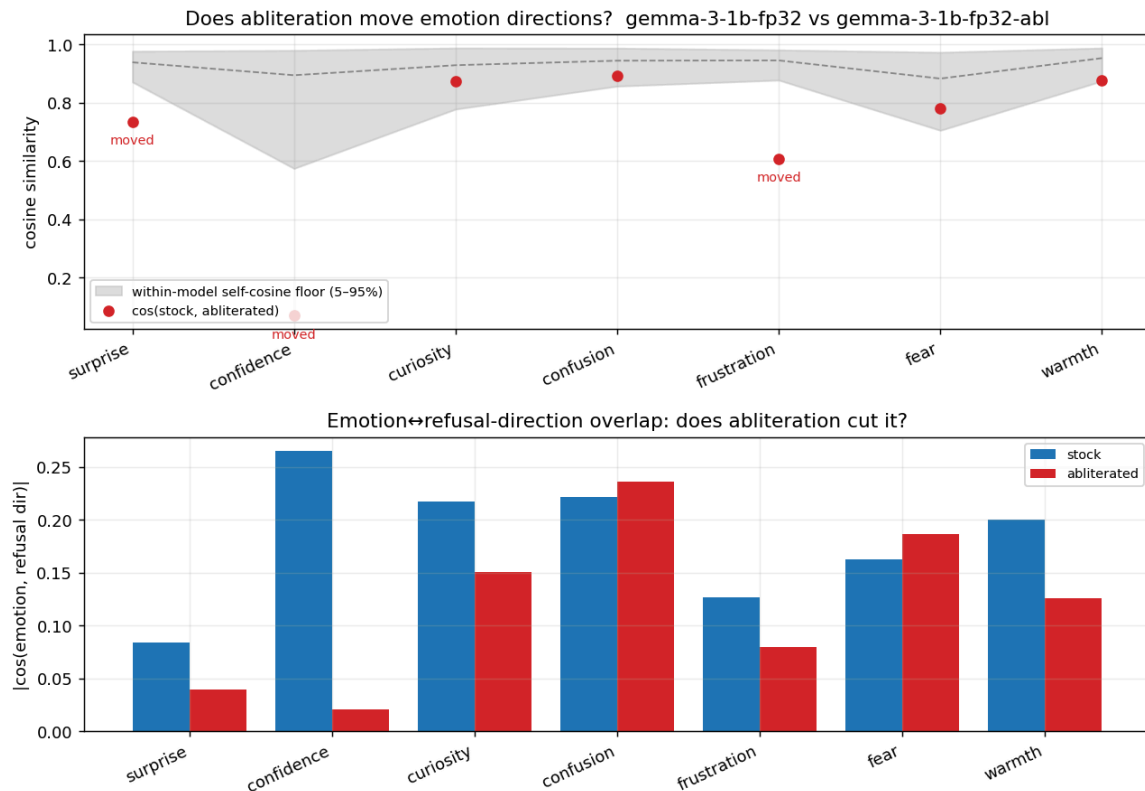


**Figure 6.** Held-out behavioural confirmation across the gemma-3 ladder. *Left:* ablation drives the harmful-prompt refusal rate to zero at every scale (AdvBench; stock in blue, ablated at 0). *Right:* it does so without inducing benign over-refusal or output degeneration.

**7.2 On the 1B model, a *partial* reorientation.** Re-fitting the emotion directions on the ablated gemma-3-1b twin (float32, which avoids the ablated model’s fp16 late-layer overflow) and comparing each to its stock counterpart against a within-model bootstrap noise floor, **three of seven** directions move beyond that floor (Table 2, Figure 3). The bipolar **confidence** axis collapses ( $\cos = 0.07$ , far below its 0.58 noise-floor 5th percentile), and **frustration** and **surprise** move as well ( $\cos = 0.61, 0.74$ ); the other four sit inside the floor. The mean cross-cosine is 0.69 — a real but modest rotation. This contrasts with an earlier, weaker estimate we built on hand-written stub prompts, which found *no* movement at 1B: with a benchmark-grade refusal direction (§7.1), even the smallest model’s emotion geometry shifts, and (as §7.5 shows) the extremes of the ladder are *not* symmetric — 1B moves, 27B does not.

**Table 2.** Abliteration effect per emotion (gemma-3-1b, float32, difference-of-means,  $B = 400$  family bootstrap, AdvBench/Alpaca refusal direction).  $\cos = \cos(\text{stock}, \text{abl})$ ; *floor* = the within-model self-cosine distribution; *moved?* is true iff  $\cos$  falls below the floor’s 5th percentile. AUC = held-out AUC. Mean  $\cos$  0.69; 3/7 axes moved.

| axis        | layer | cos   | floor mean | floor p05 | moved? | AUC stock | AUC abl |
|-------------|-------|-------|------------|-----------|--------|-----------|---------|
| surprise    | 15    | 0.736 | 0.939      | 0.872     | yes    | 0.99      | 0.93    |
| confidence  | 10    | 0.070 | 0.895      | 0.575     | yes    | 0.72      | 0.43    |
| curiosity   | 8     | 0.875 | 0.929      | 0.778     | no     | 0.78      | 0.71    |
| confusion   | 8     | 0.893 | 0.945      | 0.857     | no     | 0.77      | 0.67    |
| frustration | 13    | 0.609 | 0.946      | 0.878     | yes    | 1.00      | 0.87    |
| fear        | 8     | 0.782 | 0.883      | 0.706     | no     | 0.76      | 0.66    |
| warmth      | 11    | 0.876 | 0.954      | 0.875     | no     | 0.88      | 0.79    |



**Figure 3.** Abliteration on gemma-3-1b (fp32): 3 of 7 stock↔ablated cosines (red) fall below the within-model self-cosine band (grey) — confidence, frustration and surprise move; the emotion↔refusal overlap (lower panel) is 0.08–0.27.

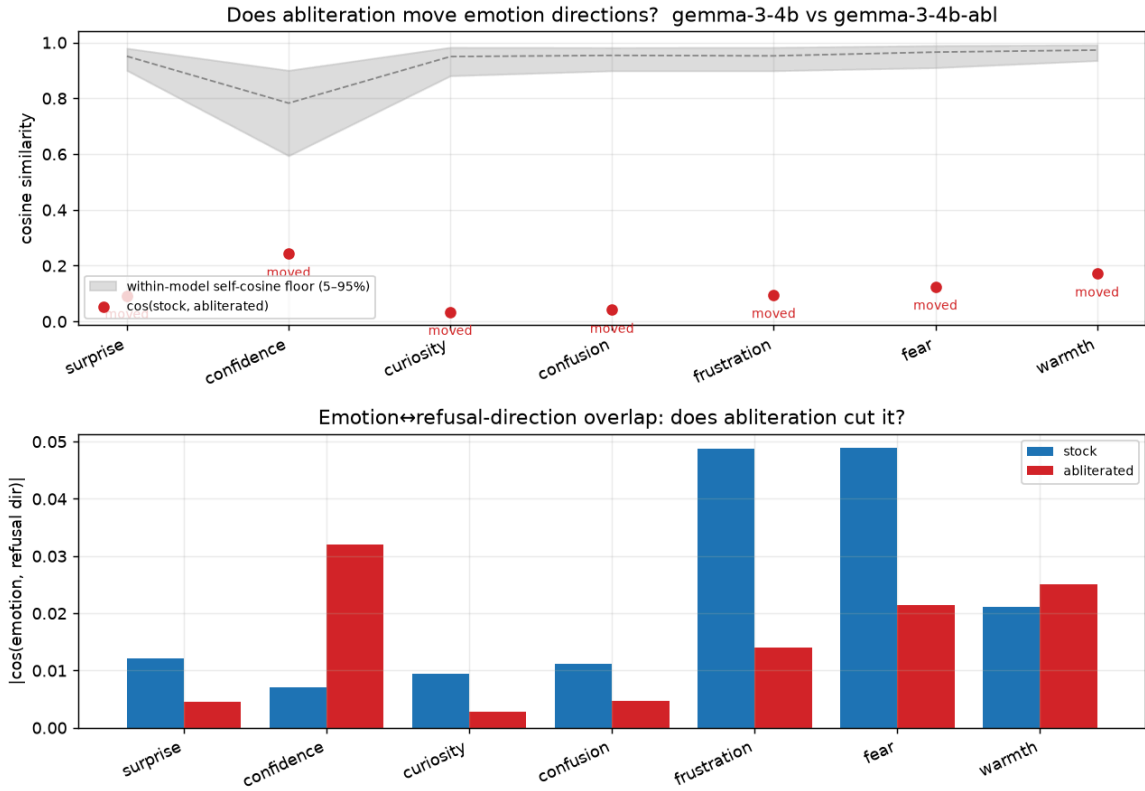
**7.3 Why it moves: small linear overlap, larger non-linear effect.** With the benchmark-grade refusal direction the emotions’ *linear* overlap with  $r$  is modest but non-trivial —  $|\cos(u_e, r)| \in [0.08, 0.27]$ , larger than the  $[0.04, 0.10]$  we measured from hand-written stubs (Figure 3, lower) — and the axis that moves most (confidence,  $\cos = 0.07$ ) also has the highest overlap (0.27). But the overlap does not *bound* the movement: ablation is a weight edit whose effect propagates non-linearly through the network, so a direction can be reorganized far more than its small projection onto  $r$  predicts. This is why the effect must be certified with a control (§7.5) rather than read off the geometry, and why the axis with negligible overlap still shifts.

**7.4 The 1B effect sits at the detection threshold.** Because the 1B rotation is modest, whether a given axis crosses the significance floor is sensitive to numerical precision: the same model shows **3/7** axes moved in float32 (Table 2) but **7/7** in bf16 (Table 3), even though the *magnitude* is stable (mean  $\cos$  0.69 vs. 0.64). We therefore treat mean cross-cosine — not the axis count — as the robust 1B quantity, and read 1B as a **partial** reorientation: real, but far weaker than the mid-scale effect in §7.5. The secondary signal is a modest separability drop on a few axes (confidence AUC 0.72→0.43, warmth 0.88→0.79) that tracks the rotation.

**7.5 Across scale: an inverted-U, and a decisive control.** Repeating the experiment on the full gemma-3 ladder in bf16 (Table 3) shows the reorientation is **strongly scale-dependent**. It peaks at the **intermediate scales** — mean  $\cos(\text{stock}, \text{abl}) = \mathbf{0.11}$  at **4B** and **0.23** at **12B**, 7/7 axes moved at both — is **partial at 1B** (0.64, §7.2–7.4), and **vanishes at 27B** (0.95, 0/7 axes moved, mean held-out AUC 0.972→0.972, *unchanged to three figures*). Throughout, the emotions remain well-separated after the edit (AUC drops  $\leq 0.10$ ): this is re-orientation of the reading direction, not destruction of the concept.

A **matched random-direction ablation control** — the *identical* weight-orthogonalization applied to a random unit direction of the same dimension — shows the effect is **refusal-specific**, not a generic consequence of editing weights (Table 3, “random”). Random ablation leaves the emotion directions essentially untouched at every scale ( $\cos$  0.95 / 0.99 / 1.00 for 1B/4B/12B, 0/7 axes moved, no meaningful AUC change, and — as expected — no reduction in refusal). So the surgery itself is inert; it is **removing the refusal direction specifically** that reorients emotions at 1B–12B.

The endpoints are the most informative. At **27B** the refusal direction is *perfectly* identifiable (AUC 1.000) and ablating it removes refusal behaviour (§7.1) — yet the emotion geometry is left **exactly** intact (AUC 0.972→0.972). The largest model represents refusal in a way that is **disentangled** from affect, so excising it really is an orthogonal edit. At **4B/12B**, refusal and emotion computation are **entangled** enough that removing refusal reorganizes the affect representation. This sharpens the safety message: “just removing refusals” is **not** a uniformly orthogonal edit — at the intermediate scales that many open “uncensored” releases occupy, it measurably restructures a model’s affective representations, and the random-direction control certifies the effect is tied to refusal rather than to weight-surgery damage. (An earlier draft, built on hand-written stub refusal data, reported the *smallest* model as “preserved” as well; with benchmark data that is corrected — 1B moves, and only 27B is genuinely preserved.)



**Figure 5.** Refusal ablation on gemma-3-4b (bf16): all seven emotion directions re-orient (red points fall well below the noise band) while separability is largely preserved — an effect the random-direction control (Table 3) shows is refusal-specific.

**Table 3.** Refusal ablation vs. a matched random-direction control (gemma-3, bf16). Low  $\text{cos}(\text{stock}, \text{abl})$  = large re-orientation; the control isolates the *refusal-specific* effect. *rr* = held-out harmful-refusal rate;  $\Delta\text{AUC}$  = held-out AUC lost (stock – ablated).

| model | rr (S→A)       | refusal cos | mv/7 | $\Delta\text{AUC}$ | random cos  | mv/7 | $\Delta\text{AUC}$ |
|-------|----------------|-------------|------|--------------------|-------------|------|--------------------|
| 1B    | 0.57 →<br>0.00 | 0.64        | 7/7  | +0.09              | 0.95        | 0/7  | +0.05              |
| 4B    | 0.23 →<br>0.00 | <b>0.11</b> | 7/7  | +0.10              | <b>0.99</b> | 0/7  | +0.01              |
| 12B   | 0.54 →<br>0.00 | <b>0.23</b> | 7/7  | +0.09              | <b>1.00</b> | 0/7  | +0.00              |
| 27B   | 0.03 →<br>0.00 | 0.95        | 0/7  | +0.00              | —           | —    | —                  |

## 8. Discussion

That refusal ablation re-orientes emotion directions at some scales but not others is informative in two ways. Practically, it bears on the safety of “uncensored” open models: a weight edit marketed as *only* removing refusals — and which we confirm behaviourally *does* remove refusal, cleanly and without collateral over-refusal, at every scale — nonetheless reshapes the

model’s affective computation at 1B–12B, and our instrument makes that measurable rather than anecdotal, with the random-direction control (§7.5) showing the effect is genuinely tied to refusal, not to the surgery. Scientifically, the scales where emotions move most (4B, 12B) are where refusal and affect are most **entangled**; that **27B** — with a *perfectly* identifiable refusal direction — is left untouched suggests the largest model has **disentangled** refusal from affect. The non-monotonicity (an inverted-U) is itself a lead for mechanistic follow-up, and a caution that a result read off any single model size need not generalize.

We stress what is and is not yet shown behaviourally. We confirm the *intervention* behaviourally — ablation drives the held-out harmful-refusal rate to zero without inducing benign over-refusal or degeneration (§7.1). But the **reorientation of the emotion directions** is so far a statement about *reading* directions (cosine), not yet about downstream affective behaviour; whether an ablated 4B/12B model’s fear or warmth *behaves* differently, not just reads differently, is the next step. This paper deliberately establishes **that** there is a measurable, refusal-specific effect before asking **what** it does.

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## 9. Limitations

- i. A 1B model has weaker, occasionally entangled emotion representations; steering has a narrow coherent range. Crucially, we **select** directions by held-out AUC (a *reading* objective) and re-use them for steering (a *causal* one); the fear axis (§6.3) reads cleanest of all seven yet does not drive the output — a read–drive dissociation traceable to a lexically-separable, user-directed probe set at an early layer. AUC therefore over-certifies: at least one of our seven axes is a topic detector rather than a behavioural handle, and read-vs-drive should be measured jointly (§10). (ii) Behavioural refusal is scored by a refusal-string detector — the field-standard convention for AdvBench, but imperfect — and stock refusal rates on AdvBench vary widely by model (0.03–0.57), so the *behavioural* headroom for ablation differs across the ladder even though the refusal *direction* is cleanly identifiable everywhere ( $\text{AUC} \geq 0.99$ ). (iii) Difference-of-means treats each emotion as one direction; genuinely multi-dimensional concepts are under-modelled. (iv) The emotion-**reorientation** result is measured on reading directions (cosine), not yet on behaviour; the matched random-direction control (§7.5) rules out a weight-surgery artifact and the *intervention* is behaviourally confirmed (§7.1), but a *behavioural* probe of the re-oriented 4B/12B models is future work. (v) The 1B reorientation sits at the detection threshold (3/7 axes in fp32, 7/7 in bf16), so we report mean cosine as the robust quantity. (vi) The bipolar confidence axis and the calibration baseline are model- and prompt-set-specific.
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## 10. Future work

**Read vs. drive (the immediate follow-up).** The fear dissociation (§6.3) motivates measuring, for every axis  $\times$  layer  $\times$  scale, **readability** (held-out AUC) against **drivability** (steering effect size on the *output*, not the meter), and selecting directions by the causal objective rather than by AUC. Axes in the high-read / low-drive corner are lexical detectors, not behavioural handles; the open questions are how many of the seven are impostors, whether corrected (model-directed, topic-matched, causally-selected) probes recover a drivable fear direction, and whether **drivability** rises with scale as **readability** does — a stronger claim than this paper’s cleanliness ladder. This doubles as a general validity check for the linear-probe literature, which routinely reports AUC alone. **Scale.** Repeat the full experiment on Gemma-4-E4B (stock vs. ablated twin) on a 24 GB GPU; the pipeline is model-agnostic and the driver is parameterized. **Controls.** The matched random-direction ablation (§7.5) establishes specificity; next is dose-response (varying strength / rank) and a *behavioural* readout of the re-oriented mid-scale models — does reorganized affect change what they do? **Discovery.** Train sparse autoencoders on both twins and ask whether the *features* — not just the named axes — shift, and whether new affective features appear. **Mechanism.** For the emotions that move, trace *what changes*: which behaviours, on which inputs, and whether the change is mediated by the removed refusal direction. This is the journey the present paper begins.

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## 11. Reproducibility statement

The toolkit, prompt sets, fitting/calibration code, steering and ablation implementations, the experiment driver, the figures, and the per-run JSON results bundles (with software versions) are released under the MIT license at [github.com/newsbubbles/faced](https://github.com/newsbubbles/faced). The ablation is a closed-form weight edit with a seeded, deterministic direction; the statistical test is a seeded bootstrap. A single command reproduces each figure from the released prompt sets (see the repository `README.md` and `WALKTHROUGH.md`). The large binary artifacts — the trained SAE and the fitted direction tensors — are regenerable from that code and are hosted separately rather than committed, so the repository stays light.

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## Figure index

Generated artifacts (regenerate with the commands in `README.md` / `WALKTHROUGH.md`):

- **Figure 1** — emotion meters over generation time: `artifacts/emotions_over_time.png`

- **Figure 2** — steering: artifacts/steering\_eval.gemma-3-1b.txt ( $\alpha$ -sweep, suppression, confusion matrix) and the peak-emotion face gallery artifacts/faces/gallery.html
- **Figure 3** — ablation effect (cosine vs. noise floor; refusal overlap), local fp32 1B: artifacts/ablation/gemma-3-1b-fp32\_vs\_gemma-3-1b-fp32-abl.png
- **Figure 4** — signal cleanliness vs. scale (AUC /  $d'$  / self-cosine heatmaps): artifacts/scaling/cleanliness.png.
- **Figure 5** — per-model ablation effect (gemma-3-4b, bf16): artifacts/ablation/gemma-3-\*\_vs\_\*-abl.png.
- **Figure 6** — held-out behavioural confirmation across the ladder (harmful-refusal  $\rightarrow$  0; benign over-refusal and degeneracy stay  $\approx$  0): artifacts/behavioral/behavioral.png.
- Raw results + provenance: artifacts/scaling/cleanliness.json, artifacts/behavioral/behavioral.json, artifacts/ablation/\*\_vs\_\*-abl.json (bf16 ladder) + the fp32 1B json, plus per-model ablation manifests. Refusal data provenance: data/refusal/SOURCES.md (AdvBench + Alpaca, disjoint fit/held-out splits).

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